

Time-Independent Perturbation Theory in Quantum Mechanics and Its Use in QFT

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Purpose and sources

Many QFT derivations use a result that first appears in ordinary quantum mechanics: if a Hamiltonian is split into a solvable part plus a small interaction, the exact energy eigenvalues and eigenstates can be expanded in powers of the interaction. This note reviews that result carefully and derives the standard second-order energy-shift formula, which is the result commonly used in QFT texts when computing interaction-induced shifts of energies or vacuum energies.

The presentation follows standard academic treatments of time-independent perturbation theory, especially Sakurai's *Modern Quantum Mechanics*, and comparable graduate-level quantum mechanics lecture notes. These sources agree on the central formulas derived below: first-order energy shifts are diagonal matrix elements of the perturbing Hamiltonian, while second-order shifts are sums over virtual intermediate unperturbed states weighted by inverse energy denominators.

1. The basic setup

Suppose the full Hamiltonian is

$$H = H_0 + \lambda V, \tag{1}$$

where H_0 is exactly solvable, V is a time-independent perturbation, and λ is a formal expansion parameter. At the end one usually sets $\lambda = 1$, but keeping λ makes the ordering of terms transparent.

The unperturbed eigenvalue problem is

$$H_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle. \tag{2}$$

We assume first that the level of interest, called $|n^{(0)}\rangle$, is non-degenerate:

$$E_m^{(0)} \neq E_n^{(0)} \quad \text{for all } m \neq n. \tag{3}$$

The exact eigenvalue equation is

$$H |n\rangle = E_n |n\rangle. \tag{4}$$

Since V is small, we expand both the exact energy and the exact eigenstate:

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + O(\lambda^3), \quad (5)$$

$$|n\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + O(\lambda^3). \quad (6)$$

The superscript (k) denotes the k th-order correction, not a power of a state.

2. Normalization convention

The exact state can be normalized in several equivalent ways. The most convenient convention for perturbation theory is intermediate normalization:

$$\langle n^{(0)} | n \rangle = 1. \quad (7)$$

Using (6), this implies

$$\langle n^{(0)} | n^{(1)} \rangle = 0, \quad \langle n^{(0)} | n^{(2)} \rangle = 0, \quad \dots \quad (8)$$

Thus all corrections to the state lie in the subspace orthogonal to the original unperturbed state. This convention removes an otherwise arbitrary component of $|n^{(k)}\rangle$ parallel to $|n^{(0)}\rangle$.

3. Expanding the Schrödinger equation order by order

Substitute (1), (5), and (6) into (4):

$$\begin{aligned} (H_0 + \lambda V) \left(|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \right) &= \left(E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \right) \\ &\times \left(|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \right). \end{aligned} \quad (9)$$

Equating equal powers of λ gives a hierarchy of equations.

At zeroth order:

$$H_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle, \quad (10)$$

which is just the unperturbed problem.

At first order:

$$H_0 |n^{(1)}\rangle + V |n^{(0)}\rangle = E_n^{(0)} |n^{(1)}\rangle + E_n^{(1)} |n^{(0)}\rangle. \quad (11)$$

Rearranging,

$$(H_0 - E_n^{(0)}) |n^{(1)}\rangle = -(V - E_n^{(1)}) |n^{(0)}\rangle. \quad (12)$$

At second order:

$$H_0 |n^{(2)}\rangle + V |n^{(1)}\rangle = E_n^{(0)} |n^{(2)}\rangle + E_n^{(1)} |n^{(1)}\rangle + E_n^{(2)} |n^{(0)}\rangle. \quad (13)$$

Equivalently,

$$(H_0 - E_n^{(0)}) |n^{(2)}\rangle = -V |n^{(1)}\rangle + E_n^{(1)} |n^{(1)}\rangle + E_n^{(2)} |n^{(0)}\rangle. \quad (14)$$

4. First-order energy shift

Take the inner product of (11) with $\langle n^{(0)}|$:

$$\langle n^{(0)}|H_0|n^{(1)}\rangle + \langle n^{(0)}|V|n^{(0)}\rangle = E_n^{(0)} \langle n^{(0)}|n^{(1)}\rangle + E_n^{(1)} \langle n^{(0)}|n^{(0)}\rangle. \quad (15)$$

Since H_0 is Hermitian and $\langle n^{(0)}|H_0 = E_n^{(0)} \langle n^{(0)}|$, the first term on the left is

$$\langle n^{(0)}|H_0|n^{(1)}\rangle = E_n^{(0)} \langle n^{(0)}|n^{(1)}\rangle. \quad (16)$$

This cancels the corresponding term on the right. Also $\langle n^{(0)}|n^{(0)}\rangle = 1$. Therefore

$$\boxed{E_n^{(1)} = \langle n^{(0)}|V|n^{(0)}\rangle.} \quad (17)$$

The physical meaning is simple: to first order, the energy shift is the expectation value of the perturbing Hamiltonian in the unperturbed state.

5. First-order correction to the state

Expand the first-order state correction in the complete unperturbed basis:

$$|n^{(1)}\rangle = \sum_m c_m^{(1)} |m^{(0)}\rangle. \quad (18)$$

Intermediate normalization gives $c_n^{(1)} = 0$, so

$$|n^{(1)}\rangle = \sum_{m \neq n} c_m^{(1)} |m^{(0)}\rangle. \quad (19)$$

Now take the inner product of (11) with $\langle m^{(0)}|$, where $m \neq n$:

$$\langle m^{(0)}|H_0|n^{(1)}\rangle + \langle m^{(0)}|V|n^{(0)}\rangle = E_n^{(0)} \langle m^{(0)}|n^{(1)}\rangle + E_n^{(1)} \langle m^{(0)}|n^{(0)}\rangle. \quad (20)$$

The final overlap is zero because $m \neq n$. Also,

$$\langle m^{(0)}|H_0|n^{(1)}\rangle = E_m^{(0)} \langle m^{(0)}|n^{(1)}\rangle. \quad (21)$$

Therefore

$$(E_m^{(0)} - E_n^{(0)}) \langle m^{(0)}|n^{(1)}\rangle = - \langle m^{(0)}|V|n^{(0)}\rangle. \quad (22)$$

Thus

$$c_m^{(1)} = \langle m^{(0)}|n^{(1)}\rangle = \frac{\langle m^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}}. \quad (23)$$

The first-order corrected state is therefore

$$\boxed{|n^{(1)}\rangle = \sum_{m \neq n} |m^{(0)}\rangle \frac{\langle m^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}}.} \quad (24)$$

This formula shows why non-degenerate perturbation theory fails near degeneracy: if $E_m^{(0)} \approx E_n^{(0)}$, the denominator becomes small and the correction may no longer be perturbative.

6. Second-order energy shift

Now derive the formula most often imported into QFT calculations. Take the inner product of (13) with $\langle n^{(0)} |$:

$$\langle n^{(0)} | H_0 | n^{(2)} \rangle + \langle n^{(0)} | V | n^{(1)} \rangle = E_n^{(0)} \langle n^{(0)} | n^{(2)} \rangle + E_n^{(1)} \langle n^{(0)} | n^{(1)} \rangle + E_n^{(2)} \langle n^{(0)} | n^{(0)} \rangle. \quad (25)$$

As before,

$$\langle n^{(0)} | H_0 | n^{(2)} \rangle = E_n^{(0)} \langle n^{(0)} | n^{(2)} \rangle, \quad (26)$$

so those terms cancel. The intermediate-normalization condition gives $\langle n^{(0)} | n^{(1)} \rangle = 0$, and $\langle n^{(0)} | n^{(0)} \rangle = 1$. Hence

$$E_n^{(2)} = \langle n^{(0)} | V | n^{(1)} \rangle. \quad (27)$$

Substitute the first-order state correction (24):

$$E_n^{(2)} = \langle n^{(0)} | V \sum_{m \neq n} | m^{(0)} \rangle \frac{\langle m^{(0)} | V | n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} \quad (28)$$

$$= \sum_{m \neq n} \frac{\langle n^{(0)} | V | m^{(0)} \rangle \langle m^{(0)} | V | n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}}. \quad (29)$$

If V is Hermitian, then

$$\langle n^{(0)} | V | m^{(0)} \rangle = \langle m^{(0)} | V | n^{(0)} \rangle^*, \quad (30)$$

and therefore

$$E_n^{(2)} = \sum_{m \neq n} \frac{|\langle m^{(0)} | V | n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}}. \quad (31)$$

This is the standard time-independent perturbation theory result that many QFT texts use. If the textbook labels this formula as equation (4.34), then (31) is the desired endpoint of the derivation.

Combining terms, the energy through second order is

$$E_n = E_n^{(0)} + \lambda \langle n^{(0)} | V | n^{(0)} \rangle + \lambda^2 \sum_{m \neq n} \frac{|\langle m^{(0)} | V | n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}} + O(\lambda^3). \quad (32)$$

7. Operator form: projection and resolvent

It is often useful, especially in field theory, to rewrite the same derivation using projectors. Define

$$P_n = |n^{(0)}\rangle \langle n^{(0)}|, \quad Q_n = 1 - P_n. \quad (33)$$

On the subspace orthogonal to $|n^{(0)}\rangle$, the operator $E_n^{(0)} - H_0$ is invertible if the level is non-degenerate. Therefore (24) can be written as

$$|n^{(1)}\rangle = \frac{Q_n}{E_n^{(0)} - H_0} V |n^{(0)}\rangle. \quad (34)$$

Then

$$E_n^{(2)} = \left\langle n^{(0)} \left| V \frac{Q_n}{E_n^{(0)} - H_0} V \right| n^{(0)} \right\rangle. \quad (35)$$

Inserting a complete set of H_0 eigenstates inside the resolvent gives back (31). This form makes the connection to QFT clearer: intermediate states propagate with an energy denominator, and matrix elements of V create and annihilate virtual excitations.

8. Degenerate perturbation theory

If the unperturbed level is degenerate, the denominator in (24) can vanish. Suppose there are g orthonormal states

$$|a^{(0)}\rangle, \quad a = 1, \dots, g, \quad (36)$$

with the same unperturbed energy $E_D^{(0)}$. Let P_D project onto this degenerate subspace. The first-order problem is obtained by projecting the Schrödinger equation into the degenerate subspace:

$$P_D V P_D |\alpha^{(0)}\rangle = E_\alpha^{(1)} |\alpha^{(0)}\rangle. \quad (37)$$

In components this is the matrix eigenvalue problem

$$\sum_{b=1}^g V_{ab} c_b^{(\alpha)} = E_\alpha^{(1)} c_a^{(\alpha)}, \quad V_{ab} = \langle a^{(0)} | V | b^{(0)} \rangle. \quad (38)$$

Thus the correct zeroth-order basis is the basis that diagonalizes V inside the degenerate subspace. Once that diagonalization is performed, the states that no longer remain degenerate can be treated with the non-degenerate formulas above, with sums over states outside the degenerate subspace.

Physically, degenerate perturbation theory explains level splitting. Examples include the Stark effect, the Zeeman effect, fine and hyperfine structure in atomic spectra, and symmetry-breaking splittings in many-body systems.

9. Interpretation of the second-order formula

The second-order shift

$$E_n^{(2)} = \sum_{m \neq n} \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}} \quad (39)$$

has three important structural features.

First, only off-diagonal matrix elements contribute. The perturbation must be able to connect the state $|n^{(0)}\rangle$ to an intermediate state $|m^{(0)}\rangle$.

Second, the numerator is positive for a Hermitian perturbation, but the sign of each contribution is controlled by the energy denominator. Intermediate states above $E_n^{(0)}$ lower the energy, while intermediate states below $E_n^{(0)}$ raise it.

Third, the formula has the form

$$\text{energy shift} \sim \frac{\text{transition amplitude} \times \text{transition amplitude}}{\text{energy mismatch}}. \quad (40)$$

This is the seed of the “virtual intermediate state” language that becomes ubiquitous in QFT.

10. How this enters quantum field theory

Quantum field theory is still quantum mechanics: it has a Hilbert space, states, operators, and a Hamiltonian. The difference is that the Hilbert space is a Fock space with variable particle number, and the Hamiltonian is usually written as

$$H = H_0 + H_{\text{int}}. \quad (41)$$

Here H_0 is the exactly solvable free-field Hamiltonian, while H_{int} contains interactions such as

$$\frac{\lambda}{4!} \int d^3x \phi^4(x), \quad e \int d^3x \bar{\psi}(x) \gamma^\mu A_\mu(x) \psi(x), \quad \int d^3x J(x) \phi(x). \quad (42)$$

Time-independent perturbation theory applies when one asks for stationary properties of the interacting Hamiltonian: energy shifts, vacuum-energy shifts, bound-state corrections, and effective low-energy Hamiltonians.

10.1 Vacuum energy shifts

Let $|0\rangle$ be the free vacuum, with unperturbed energy $E_0^{(0)}$. Treating H_{int} as V , the second-order vacuum-energy shift is

$$E_0^{(2)} = \sum_{N \neq 0} \frac{|\langle N | H_{\text{int}} | 0 \rangle|^2}{E_0^{(0)} - E_N^{(0)}}. \quad (43)$$

The sum now runs over all free-field Fock states $|N\rangle$: one-particle, two-particle, and multiparticle states, including integrals over their momenta. This is the Hamiltonian-language origin of vacuum bubble diagrams and vacuum energy corrections.

10.2 Static sources and induced potentials

If a field couples to a static external source, for example

$$H_{\text{int}} = \int d^3x J(\vec{x}) \phi(\vec{x}), \quad (44)$$

then H_{int} can connect the vacuum to one-particle intermediate states. Inserting those states into (43) produces an energy shift quadratic in the source:

$$E_0^{(2)} \sim -\frac{1}{2} \int d^3x d^3y J(\vec{x}) G(\vec{x} - \vec{y}) J(\vec{y}), \quad (45)$$

where G is the appropriate static Green's function. For a massless mediator, $G(\vec{r}) \propto 1/(4\pi r)$, giving the familiar Coulomb-type potential. For a massive mediator, $G(\vec{r}) \propto e^{-mr}/(4\pi r)$, giving the Yukawa potential. This is one of the most direct places where the quantum-mechanical formula (31) appears inside a QFT derivation.

10.3 Old-fashioned perturbation theory and energy denominators

Before covariant Feynman-diagram rules are introduced, QFT perturbation theory can be formulated using Hamiltonian perturbation theory on Fock space. In this “old-fashioned” perturbation theory, intermediate states are physical on-shell free-particle states, and each intermediate state contributes an energy denominator of the form

$$\frac{1}{E_{\text{initial}}^{(0)} - E_{\text{intermediate}}^{(0)}}. \quad (46)$$

This is exactly the same structure as the denominator in (31). Covariant perturbation theory reorganizes these terms into propagators, but the underlying physical idea is already visible in ordinary quantum mechanics.

10.4 Bound-state and spectral corrections

In QED and other field theories, many observable quantities are energy shifts of stationary states. Examples include radiative corrections to atomic energy levels, hyperfine splittings, Lamb-shift physics, and effective spin-dependent interactions. The actual calculations require renormalization and relativistic field-theoretic ingredients, but the logic is the same: split the Hamiltonian into a solvable part and a perturbation, compute matrix elements between unperturbed states, and sum over intermediate states.

10.5 Effective Hamiltonians

Time-independent perturbation theory also explains why high-energy states can induce low-energy interactions. If P projects onto a low-energy subspace and $Q = 1 - P$ projects onto high-energy states, then to second order the low-energy subspace feels an effective interaction of the schematic form

$$H_{\text{eff}}^{(2)} = PVQ \frac{1}{E - H_0} QVP. \quad (47)$$

This is the Hamiltonian counterpart of integrating out heavy fields in effective field theory. The resulting low-energy operators are suppressed by large energy or mass denominators, just as in (31).

11. Summary of the derivation chain to equation (4.34)

The chain of reasoning is:

1. Split the Hamiltonian as $H = H_0 + \lambda V$.
2. Solve the unperturbed problem $H_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle$.
3. Expand E_n and $|n\rangle$ in powers of λ .
4. Insert the expansions into the exact eigenvalue equation.
5. Equate powers of λ .
6. Project the first-order equation onto $|n^{(0)}\rangle$ to obtain

$$E_n^{(1)} = \langle n^{(0)} | V | n^{(0)} \rangle. \quad (48)$$

7. Project the first-order equation onto $|m^{(0)}\rangle$ with $m \neq n$ to obtain

$$|n^{(1)}\rangle = \sum_{m \neq n} |m^{(0)}\rangle \frac{\langle m^{(0)} | V | n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}}. \quad (49)$$

8. Project the second-order equation onto $|n^{(0)}\rangle$ to obtain

$$E_n^{(2)} = \langle n^{(0)} | V | n^{(1)} \rangle. \quad (50)$$

9. Substitute the first-order state correction to obtain

$$E_n^{(2)} = \sum_{m \neq n} \frac{|\langle m^{(0)} | V | n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}}. \quad (4.34)$$

This final expression is the key time-independent perturbation theory result used in many QFT contexts. In QFT, $|m^{(0)}\rangle$ becomes a free Fock-space intermediate state, V becomes the interaction Hamiltonian, and the sum over m becomes a sum and integral over all allowed intermediate particles and momenta.

References

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5. L. I. Schiff, *Quantum Mechanics*, 3rd ed., McGraw-Hill, standard treatment of stationary-state perturbation theory.
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